

Job Transformation, Specialization, and the Labor Market Effects of AI

Lukas Freund

Boston College

Lukas Mann

Arizona State University

May 28, 2026

AI in the Workplace Conference, University of Chicago

A framework to quantify the effects of AI-induced job transformation effects

- What will be the labor market consequences of AI?
- Our focus: **job transformation** = shift in the task content of jobs

Junior associates [...] will historically have been doing a lot of document review work [...]

A lot of that can be automated [...]

But that just means actually those associates can be used more efficiently on other things [...] It's upscaling the type of work they're able to do [like] interacting with clients and just doing more sophisticated legal work



O'Connor and Burn-Murdoch (2025, FT)

A framework to quantify the effects of AI-induced job transformation effects

- What will be the labor market consequences of AI?
- Our focus: **job transformation** = shift in the task content of jobs

⇒ **How will AI affect wages through job transformation? Winners and losers?**

- **Challenge:** *wage effects* depend on *task-specific* skills, which are *unobserved*
 - state-of-art models abstract from job transformation effects

A framework to quantify the effects of AI-induced job transformation effects

- What will be the labor market consequences of AI?
- Our focus: **job transformation** = shift in the task content of jobs

⇒ **How will AI affect wages through job transformation? Winners and losers?**

- **Challenge:** *wage effects* depend on *task-specific* skills, which are *unobserved*
→ state-of-art models abstract from job transformation effects
- **This paper:**
 - ① Theory: formalize **job transformation wage effects** in a task-based GE model
 - ② Measurement: estimate **skill distribution**
 - ③ Application: project worker-level wage effects due to **genAI**

Step 1: Formalize job transformation effects in a task-based GE model

[► Details](#)

- **Final production** requires the output of different occupations (GE)
- **Occupational production:** each occ. $o \in \mathcal{O}$ bundles tasks $\mathcal{T}$ with specific weights
- **Tasks** are assigned to labor ($\rightarrow \mathcal{T}_l$) or machines ($\rightarrow \mathcal{T}_m$)
 - *task bundling:* a worker in o has to perform *all* the tasks assigned to labor in o
 - occupational (labor) task weight matrix $A = \{A_{o,\tau}\}_{\tau \in \mathcal{T}_l, o \in \mathcal{O}}$
- **Workers** have heterogeneous, task-specific skills, $s_i \equiv \{s_{i,\tau}\}_{\tau \in \mathcal{T}_l}$
 - choose occupation to maximize utility

Model yields a tractable and intuitive log-linear wage

$$w_{i,\cdot,t} = \mu + A s_i + \varepsilon_{i,t}$$

$$\begin{aligned}
 \underbrace{\text{log wage of } i \text{ if choose } o \text{ in } t}_{w_{i,o,t}} &= \underbrace{\text{occ.-specific intercept (incl. prices)}}_{\mu_o} + \underbrace{\sum_{\mathcal{T}_l} A_{o,\tau} \cdot s_{i,\tau}}_{\text{weighted skills}} + \underbrace{\varepsilon_{i,t}}_{\text{idiosyncratic productivity shock (not crucial)}} \\
 &= \mu_o + \underbrace{\frac{1}{|\mathcal{T}_l|} \sum_{\mathcal{T}_l} s_{i,\tau}}_{\text{scalar absolute advantage } \perp o} + |\mathcal{T}_l| \cdot \underbrace{\text{Cov}(A_{o,\cdot}, s_{i,\cdot})}_{\text{comparative advantage in } o} + \varepsilon_{i,t}
 \end{aligned}$$

Given task bundling, automation leads to job transformation

- **Automation:** rise in machine productivity at some task τ^* making it optimal to reassign τ^* from labor ($\mathcal{T}_l$) to machines ($\mathcal{T}_m$)
- **Job transformation** = change in A_o : weight on τ^* $\downarrow$ & weight on other entries $\uparrow$

	τ^*	Task 2	Task 3	Task 4	Task 5
A_o (Pre)	0.2	0	0.4	0.1	0.3
A_o (Post)	0	0	0.5	0.125	0.375

- **Lawyers:** *less* weight on document review, *more* time with interactive tasks

Wages change due to canonical *and* job-transformation effects

$$\underbrace{\mathbb{E} [w_{i,o,t+1} - w_{i,o,t}]}_{\Delta \text{ potential log wage for } i \text{ in } o} = \Delta \mu_o + \underbrace{(A'_o - A_o)}_{\Delta \text{ task weights in } o} \cdot \underbrace{s_i}_{\text{individual skill vector}}$$

1 Occupation-level effects:

$$\Delta \mu_o = (\text{productivity vs. displacement effects}) + (\text{GE price effects})$$

2 Job transformation effects:

$$(A'_o - A_o) \cdot s_i = \underbrace{A_{o,\tau^*}}_{\text{occupational exposure}} \times \left(\sum_{\tau \in \mathcal{T} \setminus \tau^*} \underbrace{A'_{o,\tau} s_{i,\tau} - s_{i,\tau^*}}_{i\text{'s relative specialization}} \right)$$

Step 2: estimate the distribution of task-specific skills

→ We parametrize the model at same 'resolution' as task-exposure measures

1 Map model tasks & occupations to data, *construct A*

[► Details](#)

- tasks: NLP tools to cluster $\sim 20,000$ O*NET task statements $\rightarrow$ 38 model tasks
- time-varying occ. task weights (A): pre-2000 & post-2000

2 Estimate skill distribution

[► Details](#)

- assume lognormal skills: $s_i \equiv \{s_{i,\tau}\}_{\tau \in \mathcal{T}_i} \sim \mathcal{N}(\bar{s}, \Sigma_s) \rightarrow$ estimate $(\bar{s}, \Sigma_s)$
- data: A + NLSY panel of worker occ. choices and wages
- identifying variation: realized wages & occupational choices
- maximum likelihood estimation (MLE)

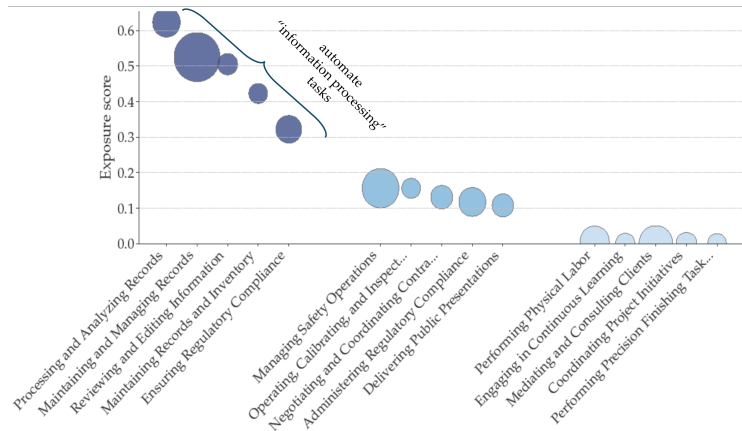
3 Validate the estimates

[► Details](#)

- fits empirical steady-state moments
- captures historical job transformation effects due to routine-biased tech. change (RBTC)

Step 3: project worker-level wage effects due to AI-induced job transformation

→ If LLMs automate certain tasks, what are the worker-level wage effects?



Share of detailed O*NET tasks within each task cluster rated as fully automatable by Eloundou et al. (2024).

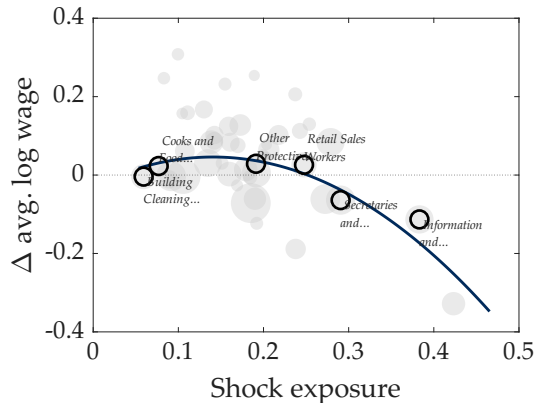
- ① Exposure: What are the implications for wages of being in an AI-exposed occupation?

Result #1: Some exposure is good, on average, but too much is bad

► Dispersion

[+] for low exposure → focus on what you're best at

[-] for high exposure → machines take over what you're best at

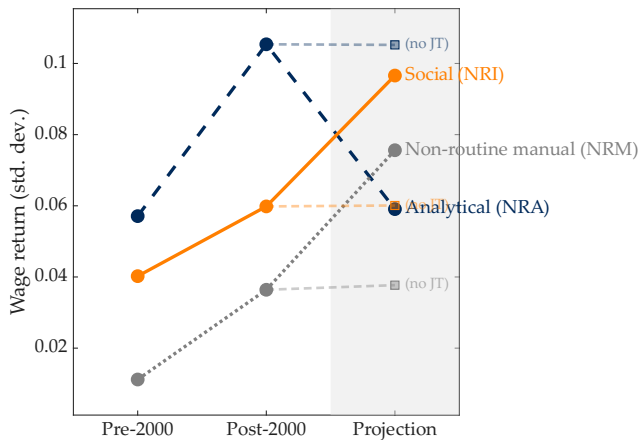


Vertical axis: change in average individual wage; horizontal axis: exposure of individual's origin occupation.

② **Skills**: What type of skills become more or less valuable due to AI?

Result #2: AI raises the return to social, lowers that of analytical skills

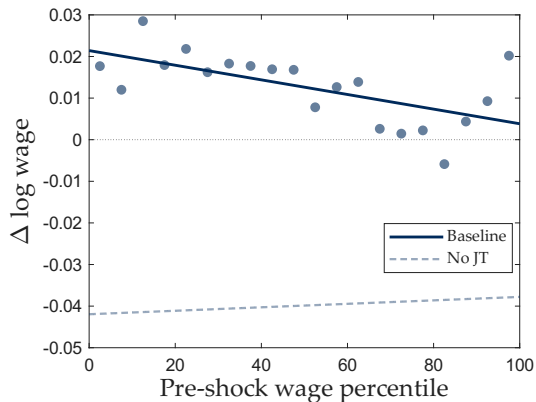
→ Job transformation drives shifts in skill returns



③ Distribution: Will high-wage or low-wage workers do better due to AI?

Result #3: AI shock is mildly “progressive” *because* of JT

→ Wage gains due to job transformation disproportionately accrue to low-wage workers



A framework to quantify the effects of AI-induced job transformation

We develop a framework to quantify job transformation effects and demonstrate their central role in shaping the labor market consequences of AI.

- ① AI **exposure** signals potential for change, w/ large heterogeneity within occupation
- ② AI “complements” social **skills** but “substitutes” for analytical skills
- ③ AI’s **distributional** impact is mildly progressive

Thank You!

Extra Slides

Theory

Model environment: task-based production meets Roy

[► Equilibrium definition](#)
[◀ Back](#)

Technology

Final good: CES (σ) aggregator over occupational output

Occupation: $o \in \mathcal{O}$ **bundles** tasks $\mathcal{T}$ with weights $\{\alpha_{o,\tau}\}_{\tau \in \mathcal{T}}$

$$Y_{i,o,t} = \prod_{\tau \in \mathcal{T}} X_{i,\tau,t}^{\alpha_{o,\tau}}$$

where $X_{i,\tau,t} = X_{i,\tau,t}^{\text{machines}} + X_{i,\tau,t}^{\text{labor}}$

Tasks: Assigned to labor ($\rightarrow \mathcal{T}_l$) or machines ($\rightarrow \mathcal{T}_m$)

Firm/job: Hires 1 worker (i), chooses machine quantity $M_{i,\tau,t}$
 $\rightarrow X_{i,\tau,t}^{\text{machines}} = \exp(z_\tau) M_{i,\tau,t}$

Model environment: task-based production meets Roy

► Equilibrium definition

◀ Back

Technology

Final good: CES (σ) aggregator over occupational output

Occupation: $o \in \mathcal{O}$ **bundles** tasks $\mathcal{T}$ with weights $\{\alpha_{o,\tau}\}_{\tau \in \mathcal{T}}$

$$Y_{i,o,t} = \prod_{\tau \in \mathcal{T}} X_{i,\tau,t}^{\alpha_{o,\tau}}$$

where $X_{i,\tau,t} = X_{i,\tau,t}^{\text{machines}} + X_{i,\tau,t}^{\text{labor}}$

Tasks: Assigned to labor ($\rightarrow \mathcal{T}_l$) or machines ($\rightarrow \mathcal{T}_m$)

Firm/job: Hires 1 worker (i), chooses machine quantity $M_{i,\tau,t}$
 $\rightarrow X_{i,\tau,t}^{\text{machines}} = \exp(z_\tau) M_{i,\tau,t}$

Workers

Skills: **Task-specific**, time-invariant

$$s_i \equiv \{s_{i,\tau}\}_{\tau \in \mathcal{T}_l} \sim \mathcal{N}(\bar{s}, \Sigma_s)$$

$$\rightarrow X_{i,\tau,t}^{\text{labor}} = \exp(s_{i,\tau}) \ell_{i,\tau,t}$$

Occupational choice:

Choose $o \in \mathcal{O}$ s.t. Gumbel preference shocks

Time: Inelastic unit supply

$$\sum_{\tau \in \mathcal{T}_l} \ell_{i,\tau,t} = 1$$

Model environment: task-based production meets Roy

► Equilibrium definition

◀ Back

Technology

Final good: CES (σ) aggregator over occupational output

Occupation: $o \in \mathcal{O}$ **bundles** tasks $\mathcal{T}$ with weights $\{\alpha_{o,\tau}\}_{\tau \in \mathcal{T}}$

$$Y_{i,o,t} = \prod_{\tau \in \mathcal{T}} X_{i,\tau,t}^{\alpha_{o,\tau}}$$

where $X_{i,\tau,t} = X_{i,\tau,t}^{\text{machines}} + X_{i,\tau,t}^{\text{labor}}$

Tasks: Assigned to labor ($\rightarrow \mathcal{T}_l$) or machines ($\rightarrow \mathcal{T}_m$)

Firm/job: Hires 1 worker (i), chooses machine quantity $M_{i,\tau,t}$
 $\rightarrow X_{i,\tau,t}^{\text{machines}} = \exp(z_\tau) M_{i,\tau,t}$

Workers

Skills: **Task-specific**, time-invariant

$$s_i \equiv \{s_{i,\tau}\}_{\tau \in \mathcal{T}_l} \sim \mathcal{N}(\bar{s}, \Sigma_s)$$

$$\rightarrow X_{i,\tau,t}^{\text{labor}} = \exp(s_{i,\tau}) \ell_{i,\tau,t}$$

Occupational choice:

Choose $o \in \mathcal{O}$ s.t. Gumbel preference shocks

Time: Inelastic unit supply

$$\sum_{\tau \in \mathcal{T}_l} \ell_{i,\tau,t} = 1$$

Markets

Labor: Competitive

Capital: Infinitely elastic supply, machines can be rented at rate r

Occupational output: Competitive

Final good: Competitively traded numeraire

Optimal time allocation is proportional to weight matrix A

► Firm problem

► Capital FOC

Firm's profit maximization problem yields:

$$\ell_{i,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} = \boxed{\frac{\alpha_{o,\tau}}{LS_o}}$$

The **occupational task-weight matrix A** summarizes relative weights attached to tasks $\tau \in \mathcal{T}_l$ across occupations $o \in \mathcal{O}$:

$$A = \begin{pmatrix} \frac{\alpha_{1,1}}{LS_1} & \frac{\alpha_{1,2}}{LS_1} & \cdots & \frac{\alpha_{1,n_{skill}}}{LS_1} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\alpha_{n_{occ},1}}{LS_{n_{occ}}} & \frac{\alpha_{n_{occ},2}}{LS_{n_{occ}}} & \cdots & \frac{\alpha_{n_{occ},n_{skill}}}{LS_{n_{occ}}} \end{pmatrix} \in \mathbb{R}^{|\mathcal{O}| \times |\mathcal{T}_l|}$$

where $LS_o = \sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}$ denotes the labor share in occupation o .

Model yields a tractable and intuitive log-linear wage equation

► Intercept term

$$w_{i,\cdot,t} = \mu + A s_i + \varepsilon_{i,t}$$

$$\begin{aligned}
 \underbrace{\text{log wage of } i}_{w_{i,o,t}} \text{ if choose } o \text{ in } t &= \underbrace{\text{occ.-specific intercept (incl. prices)}}_{\mu_o} + \underbrace{\sum_{\tau_l} A_{o,\tau} \cdot s_{i,\tau}}_{\text{weighted skills}} + \underbrace{\varepsilon_{i,t}}_{\text{idiosyncratic productivity shock (not crucial)}} \\
 &= \mu_o + \underbrace{\frac{1}{n_{\text{skill}}} \sum_{\tau_l} s_{i,\tau}}_{\text{scalar absolute advantage } \perp o} + \underbrace{\text{Cov} \left(n_{\text{skill}} \cdot A_{o,\cdot}, s_{i,\cdot} - \frac{1}{n_{\text{skill}}} \sum_{\tau_l} s_{i,\tau} \right)}_{\text{comparative advantage in } o} + \varepsilon_{i,t}
 \end{aligned}$$

Automation leads to job transformation *given* task bundling

- **Automation:** rise in machine productivity z_{τ^*} making it optimal to reassign τ^*

$$\mathcal{T}'_l = \mathcal{T}_l \setminus \tau^* \quad \mathcal{T}'_m = \mathcal{T}_m \cup \tau^*$$

- **Job transformation** = change in A_o : weight on τ^* $\downarrow$ & weight on other tasks $\uparrow$ proportional to their occupation-specific weight

	τ^*	Task 2	Task 3	Task 4	Task 5
Occ 1 (Pre)	0.2	0	0.4	0.1	0.3
Occ 1 (Post)	0	0	0.5	0.125	0.375
Occ 2 (Pre)	0	0.5	0.4	0.1	0
Occ 2 (Post)	0	0.5	0.4	0.1	0

Automation leads to job transformation *given* task bundling

- **Automation:** rise in machine productivity z_{τ^*} making it optimal to reassign τ^*

$$\mathcal{T}'_l = \mathcal{T}_l \setminus \tau^* \quad \mathcal{T}'_m = \mathcal{T}_m \cup \tau^*$$

- **Job transformation:** = change in A_o : weight on τ^* $\downarrow$ & weight on other tasks $\uparrow$ proportional to their occupation-specific weight

- “**partial automation**”: only a fraction $\zeta_{\tau^*} \in [0, 1]$ is automated

[► Details](#)

- Job transformation meaningful ($A'_o - A_o \neq 0$) if an occ. features **task bundling**:

$$|\{\tau \in \mathcal{T}_l : \alpha_{o,\tau} > 0\}| > 1$$

Wages change due to canonical *and* job-transformation effects

$$\underbrace{\mathbb{E} [w_{i,o,t+1} - w_{i,o,t}]}_{\Delta \text{ potential log wage for } i \text{ in } o} = \underbrace{\Delta \mu_o + (A'_o - A_o)}_{\Delta \text{ task weights in } o} \cdot \underbrace{s_i}_{\text{individual skill vector}}$$

1 Occupation-level effects:

$$\Delta \mu_o = \underbrace{\frac{\alpha_{o,\tau^*}}{LS_o - \alpha_{o,\tau^*}} (z_{\tau^*} - \log r + \mu_o)}_{\text{productivity \& displacement effects}} + \underbrace{\left(\frac{\log P'_o}{LS_o - \alpha_{o,\tau^*}} - \frac{\log P_o}{LS_o} \right)}_{\text{GE effects}}$$

2 Job transformation effects:

$$(A'_o - A_o) \cdot s_i = \underbrace{\frac{\alpha_{o,\tau^*}}{LS_o}}_{\text{occupational exposure}} \times \underbrace{\left(\sum_{\tau \neq \tau^*} \frac{\alpha_{o,\tau}}{LS_o - \alpha_{o,\tau^*}} s_{i,\tau} - s_{i,\tau^*} \right)}_{i\text{'s relative specialization}}$$

Firm's optimal production problem

- **Output** of firm in occ o with worker i given idiosyncratic shock $\varepsilon_{i,t} \sim \mathcal{N}(0, \varrho)$:

$$y_{i,o,t}(\cdot) = \underbrace{\prod_{\tau \in \mathcal{T}_l} (\exp(s_{i,\tau} + \varepsilon_{i,t}) \cdot \ell_{i,\tau,t})^{\alpha_{o,\tau}}}_{\text{worker-produced}} \underbrace{\prod_{\tau \in \mathcal{T}_m} (\exp(z_\tau) \cdot m_{i,\tau,t})^{\alpha_{o,\tau}}}_{\text{machine-produced}}$$

- **Profits:**

$$\pi_{i,o,t} = \max_{\{m_{i,\tau}\}_{\tau \in \mathcal{T}_m}, \{\ell_{i,\tau}\}_{\tau \in \mathcal{T}_l}} p_{o,t} y_{i,o,t}(\{\ell_{i,\tau,t}\}_{\tau \in \mathcal{T}_l}, \{m_{i,\tau,t}\}_{\tau \in \mathcal{T}_m}) - \exp(w_{i,o,t}) - r \sum_{\tau \in \mathcal{T}_m} m_{i,\tau,t}$$

$$\text{s.t. } \sum_{\tau \in \mathcal{T}_l} \ell_{i,\tau,t} = 1$$

Firm's optimal production problem

[◀ Back](#)

- **Output** of firm in occ o with worker i given idiosyncratic shock $\varepsilon_{i,t} \sim \mathcal{N}(0, \varrho)$:

$$y_{i,o,t}(\cdot) = \underbrace{\prod_{\tau \in \mathcal{T}_l} (\exp(s_{i,\tau} + \varepsilon_{i,t}) \cdot \ell_{i,\tau,t})^{\alpha_{o,\tau}}}_{\text{worker-produced}} \underbrace{\prod_{\tau \in \mathcal{T}_m} (\exp(z_\tau) \cdot m_{i,\tau,t})^{\alpha_{o,\tau}}}_{\text{machine-produced}}$$

- **Profits:**

$$\pi_{i,o,t} = \max_{\{m_{i,\tau}\}_{\tau \in \mathcal{T}_m}, \{\ell_{i,\tau}\}_{\tau \in \mathcal{T}_l}} p_{o,t} y_{i,o,t}(\{\ell_{i,\tau,t}\}_{\tau \in \mathcal{T}_l}, \{m_{i,\tau,t}\}_{\tau \in \mathcal{T}_m}) - \exp(w_{i,o,t}) - r \sum_{\tau \in \mathcal{T}_m} m_{i,\tau,t}$$

$$\text{s.t. } \sum_{\tau \in \mathcal{T}_l} \ell_{i,\tau,t} = 1$$

- **Optimality:**

[▶ FOC capital](#)

$$\ell_{i,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}}$$

Firm's optimal production problem

◀ Back

- **Output** of firm in occ o with worker i given idiosyncratic shock $\varepsilon_{i,t} \sim \mathcal{N}(0, \varrho)$:

$$y_{i,o,t}(\cdot) = \underbrace{\prod_{\tau \in \mathcal{T}_l} (\exp(s_{i,\tau} + \varepsilon_{i,t}) \cdot \ell_{i,\tau,t})^{\alpha_{o,\tau}}}_{\text{worker-produced}} \underbrace{\prod_{\tau \in \mathcal{T}_m} (\exp(z_\tau) \cdot m_{i,\tau,t})^{\alpha_{o,\tau}}}_{\text{machine-produced}}$$

- **Profits:**

$$\pi_{i,o,t} = \max_{\{m_{i,\tau}\}_{\tau \in \mathcal{T}_m}, \{\ell_{i,\tau}\}_{\tau \in \mathcal{T}_l}} p_{o,t} y_{i,o,t}(\{\ell_{i,\tau,t}\}_{\tau \in \mathcal{T}_l}, \{m_{i,\tau,t}\}_{\tau \in \mathcal{T}_m}) - \exp(w_{i,o,t}) - r \sum_{\tau \in \mathcal{T}_m} m_{i,\tau,t}$$

$$\text{s.t. } \sum_{\tau \in \mathcal{T}_l} \ell_{i,\tau,t} = 1$$

- **Optimality:**

▶ FOC capital

$$\ell_{i,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \longrightarrow \text{matrix A: } |\mathcal{O}| \times |\mathcal{T}_l|$$

Capital FOC and production

- FOC for machines $M_{i,o,t} := \sum_{\tau \in \mathcal{T}_m} M_{i,o,\tau,t}$:

$$\left(\sum_{\tau \in \mathcal{T}_m} \alpha_{o,\tau} \right) \frac{p_{o,t} Y_{i,o,t}}{r} = M_{i,o,t}$$

and

$$M_{i,o,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_m} \alpha_{o,\tau}} M_{i,o,t}$$

- Plugging both FOCs into the production function yields

$$\begin{aligned} y_{i,o,t} = \log Y_{i,o,t} = & \left[\sum_{\tau \in \mathcal{T}_l} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} s_{i,\tau} \right] + \varepsilon_{i,o} + \frac{1}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \log P_{o,t} \\ & + \left[\sum_{\tau \in \mathcal{T}} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \log(\alpha_{o,\tau}) \right] - \log \left(\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau} \right) + \left[\sum_{\tau \in \mathcal{T}_m} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} (z_\tau - \log r) \right] \end{aligned}$$

Equilibrium

Remark: Equilibrium

An equilibrium is defined as a vector of prices $\vec{p}$ and a joint distribution G of occupation choices, log wages w , log skills s and idiosyncratic productivity shocks ε , such that:

- 1 all firms make zero profits, i.e., at any point in the distribution:

$$w_{i,o,t} = \mu_o + \sum_{\tau_l} \frac{\alpha_{o,\tau}}{LS_o} \cdot s_{i,\tau} + \varepsilon_{i,t}, \quad Y \equiv \left(\sum_{o \in \mathcal{O}} \omega_o^{\frac{1}{\sigma}} Y_o^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} = \sum_{o \in \mathcal{O}} p_o Y_o$$

- 2 workers optimize, i.e., the marginal distribution of occupations conditional on wages follows

$$P(\hat{o} = o | w_{i,\cdot}) = \frac{\exp(w_{i,o}/\nu)}{\sum_{o'} \exp(w_{i,o'}/\nu)}$$

- 3 the unconditional marginal distributions of skills s and occupational shocks ε follow $\mathcal{N}(\bar{s}, \Sigma_s)$ and $\mathcal{N}(0, \varsigma^2 I)$, respectively.

Wage equation: details

- Intercept

$$\mu_o = \sum_{\tau \in \mathcal{T}} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \log(\alpha_{o,\tau}) + \left(\sum_{\tau \in \mathcal{T}_m} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} (z_\tau - \log r) \right) + \frac{1}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \log P_{o,t}$$

where $P_{o,t}$ is the price of output of occupation o

- We assume that in the initial steady state there is only one composite machine task with productivity normalized to $\log r$

Partial Automation

- Our framework nests the concept of **partial automation**
 → only a fraction $\zeta_{\tau^*} \in [0, 1]$ of task τ^* can be automated
- **Modeling:** Equivalently, rewrite pre-automation production technology as

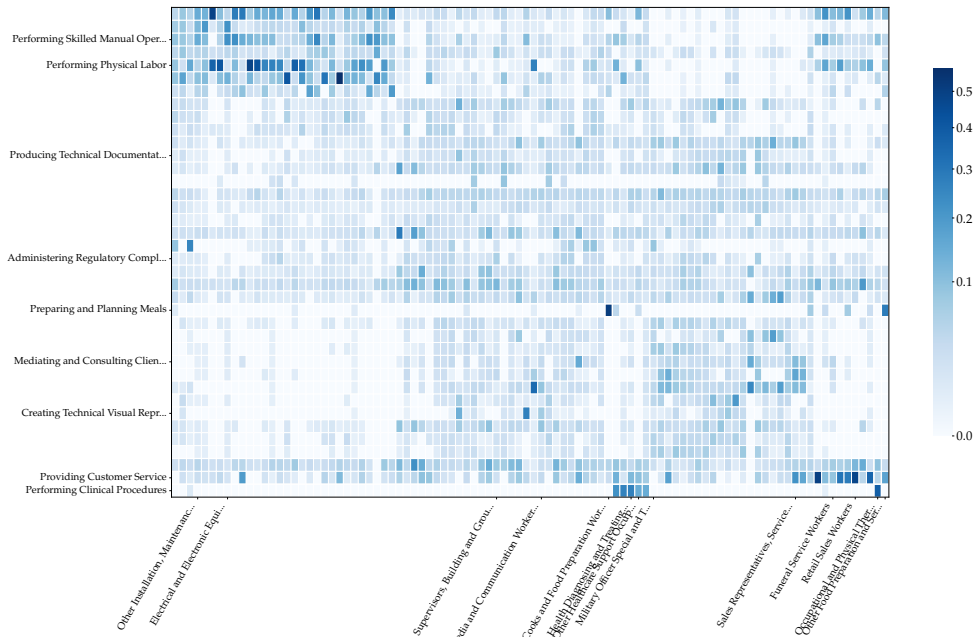
$$Y_{i,o,t} = \Gamma \cdot \prod_{\tau \in \mathcal{T}_m} (X_{i,\tau,t}^{\text{machines}})^{\alpha_{o,\tau}} \cdot \prod_{\tau \in \mathcal{T}_\ell \setminus \{\tau^*\}} (X_{i,\tau,t}^{\text{labor}})^{\alpha_{o,\tau}} \cdot ((1 - \zeta_{\tau^*}) X_{i,\tau^*,t}^{\text{labor}})^{(1 - \zeta_{\tau^*}) \alpha_{o,\tau^*}} \cdot (\zeta_{\tau^*} X_{i,\tau^*,t}^{\text{labor}})^{\zeta_{\tau^*} \alpha_{o,\tau^*}}.$$

where $\Gamma = (1 - \zeta_{\tau^*})^{-\alpha_{o,\tau^*}} (1 - \zeta_{\tau^*})^{\zeta_{\tau^*} \alpha_{o,\tau^*}}$

- Post-automation production technology becomes

$$Y_{i,o,t} = \Gamma \cdot \prod_{\tau \in \mathcal{T}_\ell \setminus \{\tau^*\}} (X_{i,\tau,t}^{\text{labor}})^{\alpha_{o,\tau}} \cdot (X_{i,\tau^*,t}^{\text{labor}})^{(1 - \zeta_{\tau^*}) \alpha_{o,\tau^*}} \cdot (X_{i,\tau^*,t}^{\text{machines}})^{\zeta_{\tau^*} \alpha_{o,\tau^*}} \cdot \prod_{\tau \in \mathcal{T}_m} (X_{i,\tau,t}^{\text{machines}})^{\alpha_{o,\tau}}.$$

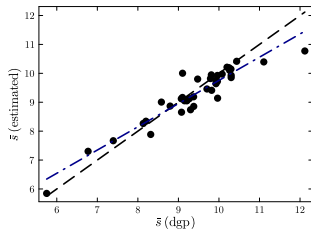
Measurement



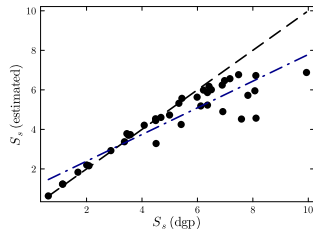
Validation: Monte-Carlo study

[◀ Back](#)

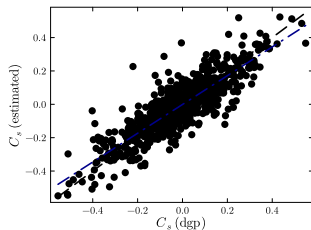
(a) Means



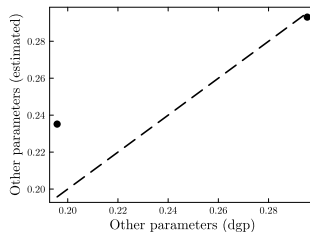
(b) Standard deviation



(c) Correlation



(d) Other parameters



Model properties & validation: overview

- Parameter estimates
 - Skill correlations vs. literature
 - Skills along the wage distribution vs. literature
- Steady state
 - Occ. employment shares
 - Wages
 - Occ. level averages
 - Between-within occ. variance decomposition
 - Occ. transitions:
 - Staying and switching probabilities
 - Switches explained by task distances
 - Individual switching frequency shaped by specialization
- Historical: RBTC
 - Skill returns: $\uparrow$ return to social skills [Deming, 2017]
 - Employment: polarization [Autor & Dorn, 2013]

Application

Result #1: Occupational exposure signals *potential* for change

[◀ Main](#)

→ Large dispersion in individual outcomes even within exposed occupations

